

High-Efficiency Ultra-Wideband Dipole Rectenna for Ambient Energy Harvesting

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Abstract—This paper designs an ultra-wideband dipole rectenna capable of ensuring high conversion efficiency at low power levels, intended for wireless power transmission and ambient energy harvesting. The receiving antenna employs a printed dipole antenna coupled to the substrate via a feed. Through meticulous design of the dipole antenna, ultra-wideband characteristics are achieved using a simple planar structure. The rectifier employs a harmonic control network to maintain high rectification efficiency across the ultra-wideband frequency range corresponding to the receiving antenna. Measurements indicate that this rectenna achieves over 50% rectification efficiency within the 1.1–3.1 GHz band at low RF power levels of 0 dBm. The planar structure of the ultra-wideband antenna facilitates integration with compact electronic devices while enabling power supply through ambient energy harvesting technology.

Keywords—dipole, energy harvesting, rectenna, ultra-wideband, wireless power transmission

I. INTRODUCTION

With the rapid development of modern wireless communication technologies, mobile networks have become increasingly widespread and dense. Consequently, the ambient radio frequency (RF) power density in the environment has significantly increased, with most of the electromagnetic spectrum concentrated below 3 GHz in the microwave band. To better capture this dispersed electromagnetic energy, ambient RF energy harvesting technology has attracted growing attention [1][2][3]. It enables direct power supply to low-power electronic devices, reducing the dependence on wired connections and battery constraints, and thereby facilitating the realization of passive sensor networks [4][5].

In ambient energy harvesting scenarios, it is essential to receive RF energy across multiple frequency bands under low ambient power densities. Therefore, wideband and low-power rectennas and rectifiers are indispensable components of energy harvesting systems [6]. In [7], a broadband rectenna with a reconfigurable matching stub was proposed. However, the use of varactor diodes increased the design complexity. Subsequently, [8] introduced an ultra-wideband rectenna

employing a novel complementary broadband matching stub, achieving over 50% efficiency under a low input power of 3 dBm across a wide frequency range. The LC resonant circuit has also been shown to provide effective conjugate matching with the antenna across a broad bandwidth, thereby eliminating antenna reactance [9].

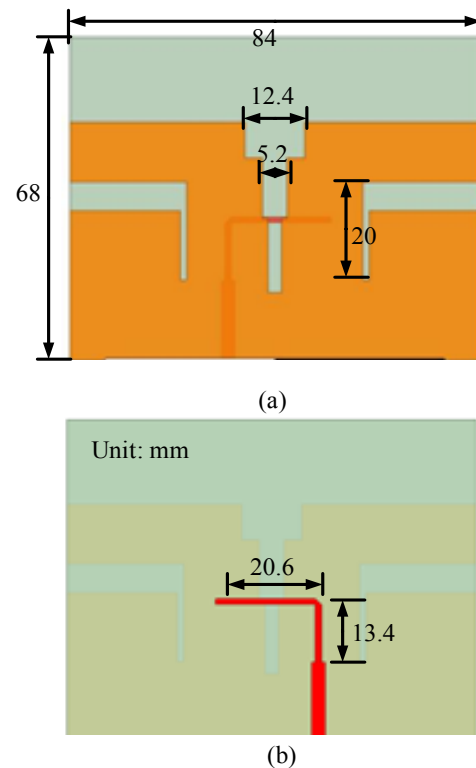


Fig. 1. Ultra-wideband dipole antenna. (a) Frontal radiator, (b) Rear feed.

In terms of rectifier design, [10] improved efficiency by compressing the input impedance, achieving over 70% efficiency across a wide frequency band, although at relatively high input power levels. [11] proposed a segmented matching

network that maintained over 50% efficiency across second frequency ranges even at a low input power of 0 dBm.

In this work, we design an ultra-wideband (UWB) dipole antenna that achieves both an extended operating bandwidth and a wide beamwidth, allowing for efficient reception of incoming waves from multiple directions. Building upon our previous work [12], we further propose an ultra-wideband rectifier based on a harmonic control network (HCN), which achieves an efficiency of over 50% at an input power of 0 dBm. The proposed antenna and rectifier exhibit well-matched bandwidths, enabling efficient ambient energy harvesting over an ultra-wide frequency range.

II. ULTRA-WIDE BAND RECTENNA DESIGN

The proposed ultra-wideband dipole rectenna comprises an ultra-wideband dipole antenna and an ultra-wideband rectenna based on a harmonic control network. Their operational bandwidths are matched to jointly achieve high efficiency in converting radio frequency to direct current across an ultra-wideband range. Both the antenna and rectifier utilize the same dielectric substrate material, with F4B-2 substrate (dielectric constant of 2.65, and loss tangent of 0.0012).

A. Ultra-wide dipole antenna design

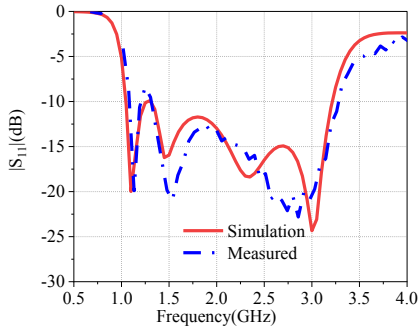


Fig. 2. Simulation and test bandwidth of ultra-wideband dipole antennas.

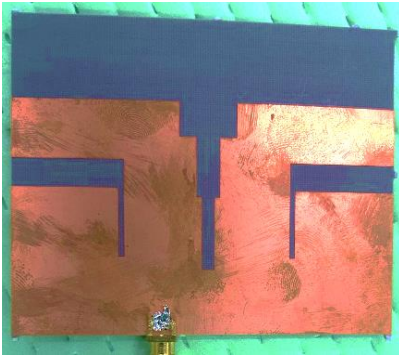
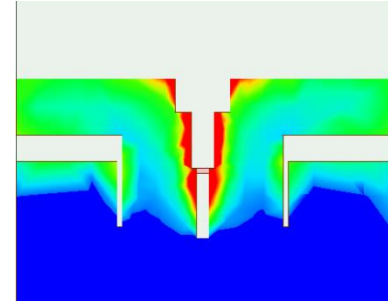


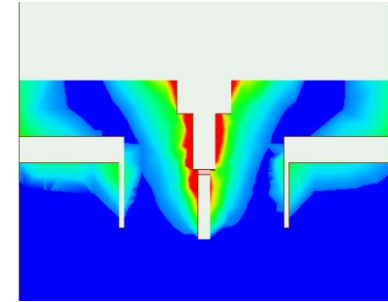
Fig. 3. Photograph of the antenna.

The dipole antenna is one of the most commonly used antenna structures. Owing to its quasi-omnidirectional radiation characteristics, it can receive electromagnetic signals from multiple directions, making it suitable for ambient energy harvesting applications. However, the limited impedance

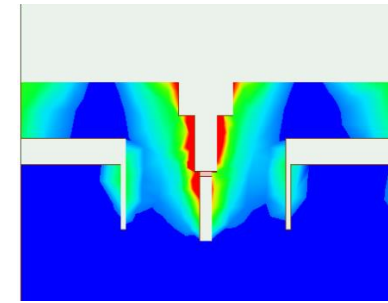
bandwidth and the feeding structure are the main challenges that restrict its further integration and performance.



(a)



(b)



(c)

Fig. 4. Antenna electric field distribution. (a) 1.3 GHz, (b) 2.4 GHz, (c) 3.0 GHz.

To extend the application of dipole antennas and enable seamless integration with electronic devices for energy supply purposes, a compact ultra-wideband microstrip dipole antenna is designed. The proposed antenna exhibits an ultra-wide impedance bandwidth and a wide beam coverage, and it can be effectively conjugate-matched with the wideband rectifier to receive ambient electromagnetic energy and power low-power electronic devices.

The front view of the designed antenna is shown in Fig. 1 (a). A coupled-feed structure is adopted, as illustrated in Fig. 1 (b), to improve impedance matching and enhance bandwidth. The antenna is fabricated on a dielectric substrate with a thickness of 1 mm. By optimizing the antenna parameters, the operating bandwidth is significantly extended. ANSYS Electronics Desktop is used to analyze the antenna performance. The simulated reflection coefficient ($|S_{11}|$) is shown in Fig. 2, demonstrating that $|S_{11}|$ remains below -10 dB over a wide frequency range from 1.1 GHz to 3.2 GHz. The proposed

antenna was fabricated, and its photograph is shown in Fig. 3. The operating bandwidth was measured using a vector network analyzer (VNA), as illustrated in Fig. 2. The measured reflection coefficient results are in good agreement with the simulated ones, validating the ultra-wideband performance of the designed antenna.

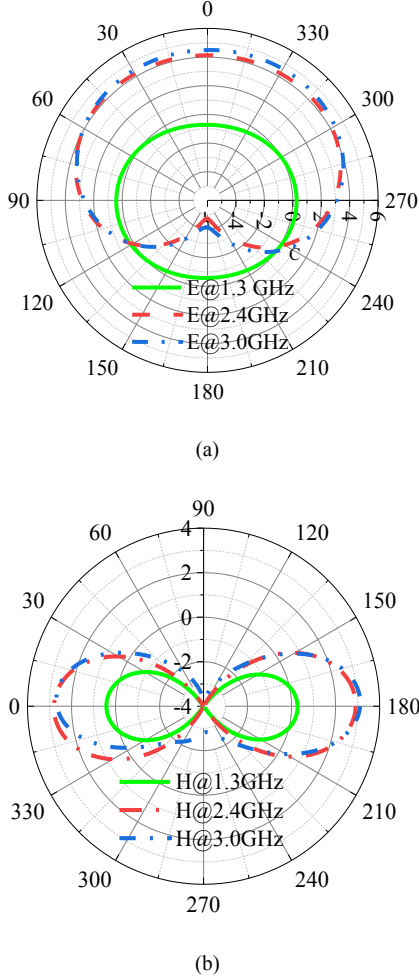


Fig. 5. Antenna radiation patterns at different frequencies. (a) E-plane, (b) H-plane.

TABLE I. SPICE PARAMETERS OF A SMS7630 DIODE

B_v	C_{j0}	I_s	R_s	V_{bi}
2.0 V	0.14 pF	5 μ A	20 Ω	0.34 V

The electric field distributions at different frequencies (1.3 GHz, 2.4 GHz, 3 GHz) are shown in Fig. 4 to illustrate the radiation behavior. The radiation patterns at representative frequencies are also presented in Fig. 5, showing stable directional characteristics across the operating band. At 1.3 GHz, where the electromagnetic wavelength is relatively long, the influence of the ground plane on the antenna performance is minimal, and the radiation pattern exhibits the typical characteristics of a standard dipole antenna. As the frequency increases, the effect of the ground plane becomes more significant. The electromagnetic waves radiated toward the

ground are partially reflected, resulting in an increase in the antenna gain by approximately 3 dB, while the beam coverage area is reduced by half. This behavior is consistent with the electric field distribution and radiation pattern.

B. The Design of Ultra-wide Rectifier

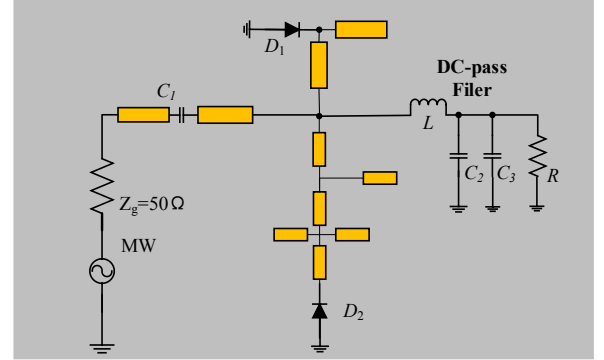


Fig. 6. Schematic diagram of an ultra-wideband rectifier.

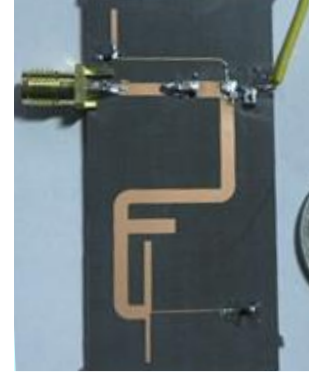


Fig. 7. Photograph of the manufactured ultra-wideband rectifier.

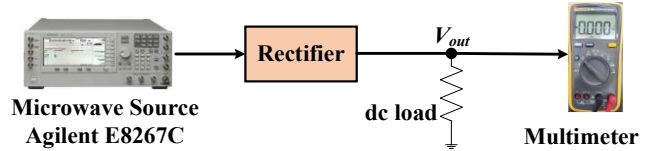


Fig. 8. Diagram of the measurement system.

In our previous research [12], we investigated rectifiers based on harmonic control networks (HCNs), which effectively enhance the rectification efficiency. Building upon this theory, an ultra-wideband rectifier was designed, as shown schematically in Fig. 6. Two parallel Schottky diodes operate at different frequency bands, ensuring high conversion efficiency across an ultra-wideband frequency range.

A lower-power rectifier based on the theoretical design was also fabricated and tested. It utilizes the low-power Schottky diode SMS7630, with its SPICE parameters listed in Table I. The fabricated prototype is shown in Fig. 7. Fig. 8 illustrates the measurement setup for rectifier efficiency. The microwave to DC conversion efficiency, denoted as η_{MW-dc} , is:

$$\eta_{MW-dc} = \frac{P_{dc}}{P_{IN}} \times 100\% = \frac{V_{OUT}^2}{R_{dc}} \times \frac{1}{P_{IN}} \times 100\% \quad (1)$$

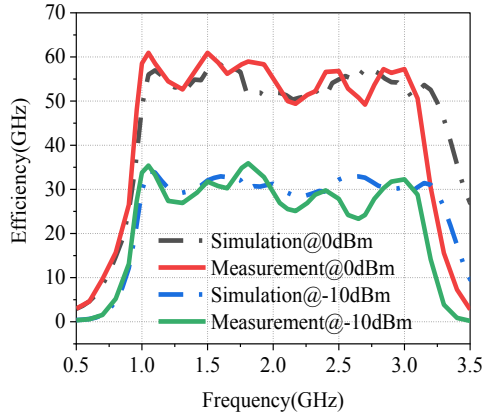


Fig. 9. Efficiency of the ultra-wideband rectifier at different power levels.



Fig. 10. Photograph of the antenna and rectifier connection.

Where V_{OUT} is the output DC voltage, and P_{IN} is the input microwave power.

Its corresponding measurement results are shown in Fig. 9. The rectifier achieves a power conversion efficiency exceeding 50% within the 1–3.1 GHz band at an input power of 0 dBm. Even when the input power decreases to –10 dBm, the rectifier still maintains an efficiency higher than 25% across the same bandwidth.

Moreover, the implementation of the HCN not only improves efficiency through higher-order harmonic tuning but also effectively compresses the impedance variation of the diodes over the entire operating bandwidth. The diode impedance after cascading the HCN is confined near the center of the Smith chart, indicating a well-matched condition. Consequently, a simple series microstrip line can be employed to optimize the input impedance of the rectifier and achieve efficient broadband impedance matching. This design concept is consistent with the feed structure of the proposed ultra-wideband dipole antenna. In this work, the antenna and rectifier are connected via an SMA connector, as shown in Fig. 10. In future developments, the antenna and rectifier can be co-designed and directly integrated, eliminating the need for an external matching network, which will further improve system compactness and integration.

III. CONCLUSION

In this paper, an ultra-wideband dipole rectenna is proposed, integrating both the wideband antenna and the rectifier. Through optimization of both components, the proposed rectenna achieves high RF-to-DC conversion efficiency at low input power levels across the 1.1–3.2 GHz frequency band. However, there remains significant potential for improvement in the antenna-rectifier interface. Future work will leverage the impedance compression capability of the HCN to achieve direct impedance matching between the antenna and rectifier, minimizing matching losses.

The proposed ultra-wideband rectifier exhibits superior broadband performance and maintains high efficiency even under low-power conditions, showing great potential for applications in ambient RF energy harvesting and wireless power transmission systems.

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